B. Tech in CSE (AI) **(SESSION- 2024-28 onwards)**

Course Code- 151301P Digital Electronics Lab 0 0 2 1

Perform any ten Experiments

List of Experiments:

1. Universal Gates (i) Identification and verification of NAND gate (IC #7400) and NOR gate (IC #7402). (ii) Construction and Verification of all other gate (AND, OR, NOT, XOR) USING a) Only NAND gate b) Only NOR gate
2. Code Convertor & Parity Generator and checker. (i) Identification & verification of NOT (7404), AND (7408) OR (7432) & XOR (7486) gates. (ii) Design, construction and verification of 3-bit Binary to Gray convertor and 3-bit Grey to Binary convertor circuit. (iii) Design, construction and verification of 3-bit odd/even Parity Generator and 4-bit odd/even parity checker circuit.
3. Adder, Subtractor & Magnitude comparator circuits. (i) Design, construction and verification of Half Adder and Half Subtractor circuit. (ii) Design, construction and verification of Full Adder and Full Subtractor circuit. (iii) Design, construction and verification of 1-bit and 4-bit Magnitude comparator. (iv) BCD Adder/Subtractor
4. Decoder, MUX & DMUX (i) Construction and verification of BCD to 7-segment decoder using IC # 7447 (ii) Verification of 4:1 MUX, 8:1 MUX & 16:1 MUX.
   (iii) Verification of 1:4 DMUX, 1:8 DMUX (iv) Cascading of MUIX and Cascading of Decoders.
5. Latches and Flip Flops (i)Construction and Verification of a Latch circuit using NAND/NOR gates. (ii) Construction and Verification of S-R Flip Flop using above Latch circuits. (iii) Verification of J-K Flip Flop using IC # 7476 (Dual J-KFF) (iv) Construction and Verification of D-Flip Flop and T-Flip Flop using J-K FF (IC #7476). (v) Construction and Verification of Master Slave J-K Flip Flop.
6. Shift Registers (i) Verification of D-FF using IC # 7474 (Dual D- FF). (ii) Construction and verification of a 2-bit Shift Right Register using IC # 7474 (iii) Construction and verification of a 2-bit Shift Left Register using IC # 7474 (iv) Verification of SISO, SIPO, PISO & PIPO Shift Registers.
7. Synchronous & Asynchronous Counters (i) Construction and verification of 2-bit Ripple counter using J-K FF. (ii) Construction and verification of Mod-3 up and Mod-3 down synchronous counter. (iii) Construction and verification of 2-bit Ring counter using J-K FF. (iv) Construction and verification of 2-bit twisted Ring (Johnson) counter using J-K FF.
8. Design and construction of a 4 bit sequence generator
9. Digital to Analog Converter (DAC). Construction & Verification of D/A converters using following methods.
   a) Weighted Resistor type
   b) R-2R ladder network type
10. Analog to Digital Converter (ADC). Construction & Verification of A/D Converter using following methods.
   a) Counter type
   b) Successive Approximation type
11. Familiarization with multisim. Design of various Digital Circuits using Multisim.